

# CS 171 / EE 142: Intro to ML and DM

Christian Shelton

UC Riverside

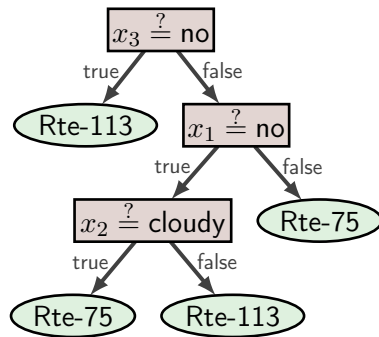
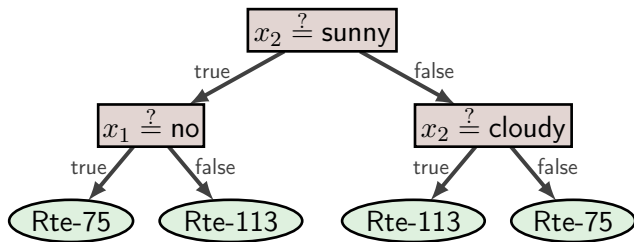
Slide Set 18: Decision Trees



- From UC Riverside
  - ▶ CS 171 / EE 142: Introduction to Machine Learning and Data Mining
  - ▶ Professor Christian Shelton
- DO NOT REDISTRIBUTE
  - ▶ For use only by enrolled students in the course

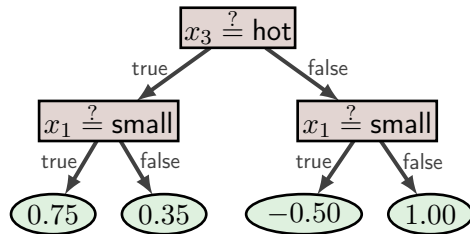
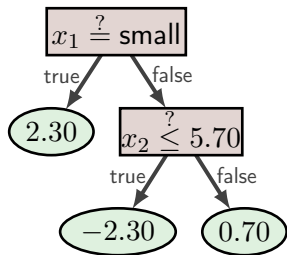
# What is a Decision Tree?

Examples of classification trees



# What is a Decision Tree?

Examples of regression trees



(which we won't cover)

# Decision Tree Hypothesis Space

The hypothesis space is all decision trees.

There are many of them.

If all features are binary, and the label is binary, there are...

# Decision Tree Hypothesis Space

The hypothesis space is all decision trees.

There are many of them.

If all features are binary, and the label is binary, there are...

$$2^{2^n}$$

different unique decision trees!

- We cannot search them all when learning.
- There are no gradients to guide search.
- We will learn (aka grow) them greedily.

# Decision Trees Split the Training Set

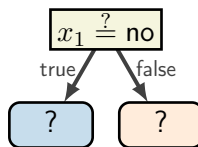
Picking the root node:

| $x_1$ | $x_2$   | $x_3$ | $y$     |
|-------|---------|-------|---------|
| no    | sunny   | no    | Rte-113 |
| no    | sunny   | yes   | Rte-113 |
| no    | cloudy  | yes   | Rte-75  |
| yes   | sunny   | no    | Rte-113 |
| no    | raining | no    | Rte-113 |
| no    | raining | yes   | Rte-75  |
| yes   | cloudy  | yes   | Rte-75  |
| no    | sunny   | no    | Rte-113 |
| yes   | sunny   | no    | Rte-75  |
| no    | cloudy  | no    | Rte-113 |

# Decision Trees Split the Training Set

Picking the root node:

| $x_1$ | $x_2$   | $x_3$ | $y$     |
|-------|---------|-------|---------|
| no    | sunny   | no    | Rte-113 |
| no    | sunny   | yes   | Rte-113 |
| no    | cloudy  | yes   | Rte-75  |
| yes   | sunny   | no    | Rte-113 |
| no    | raining | no    | Rte-113 |
| no    | raining | yes   | Rte-75  |
| yes   | cloudy  | yes   | Rte-75  |
| no    | sunny   | no    | Rte-113 |
| yes   | sunny   | no    | Rte-75  |
| no    | cloudy  | no    | Rte-113 |

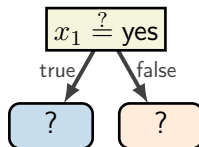




# Decision Trees Split the Training Set

Picking the root node:

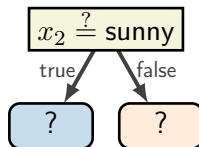
| $x_1$ | $x_2$   | $x_3$ | $y$     |
|-------|---------|-------|---------|
| no    | sunny   | no    | Rte-113 |
| no    | sunny   | yes   | Rte-113 |
| no    | cloudy  | yes   | Rte-75  |
| yes   | sunny   | no    | Rte-113 |
| no    | raining | no    | Rte-113 |
| no    | raining | yes   | Rte-75  |
| yes   | cloudy  | yes   | Rte-75  |
| no    | sunny   | no    | Rte-113 |
| yes   | sunny   | no    | Rte-75  |
| no    | cloudy  | no    | Rte-113 |



# Decision Trees Split the Training Set

Picking the root node:

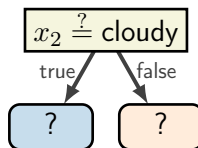
| $x_1$ | $x_2$   | $x_3$ | $y$     |
|-------|---------|-------|---------|
| no    | sunny   | no    | Rte-113 |
| no    | sunny   | yes   | Rte-113 |
| no    | cloudy  | yes   | Rte-75  |
| yes   | sunny   | no    | Rte-113 |
| no    | raining | no    | Rte-113 |
| no    | raining | yes   | Rte-75  |
| yes   | cloudy  | yes   | Rte-75  |
| no    | sunny   | no    | Rte-113 |
| yes   | sunny   | no    | Rte-75  |
| no    | cloudy  | no    | Rte-113 |



# Decision Trees Split the Training Set

Picking the root node:

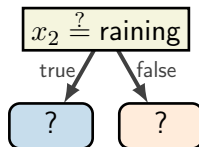
| $x_1$ | $x_2$   | $x_3$ | $y$     |
|-------|---------|-------|---------|
| no    | sunny   | no    | Rte-113 |
| no    | sunny   | yes   | Rte-113 |
| no    | cloudy  | yes   | Rte-75  |
| yes   | sunny   | no    | Rte-113 |
| no    | raining | no    | Rte-113 |
| no    | raining | yes   | Rte-75  |
| yes   | cloudy  | yes   | Rte-75  |
| no    | sunny   | no    | Rte-113 |
| yes   | sunny   | no    | Rte-75  |
| no    | cloudy  | no    | Rte-113 |



# Decision Trees Split the Training Set

Picking the root node:

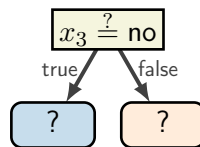
| $x_1$ | $x_2$   | $x_3$ | $y$     |
|-------|---------|-------|---------|
| no    | sunny   | no    | Rte-113 |
| no    | sunny   | yes   | Rte-113 |
| no    | cloudy  | yes   | Rte-75  |
| yes   | sunny   | no    | Rte-113 |
| no    | raining | no    | Rte-113 |
| no    | raining | yes   | Rte-75  |
| yes   | cloudy  | yes   | Rte-75  |
| no    | sunny   | no    | Rte-113 |
| yes   | sunny   | no    | Rte-75  |
| no    | cloudy  | no    | Rte-113 |



# Decision Trees Split the Training Set

Picking the root node:

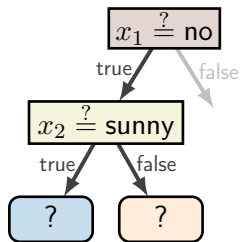
| $x_1$ | $x_2$   | $x_3$ | $y$     |
|-------|---------|-------|---------|
| no    | sunny   | no    | Rte-113 |
| no    | sunny   | yes   | Rte-113 |
| no    | cloudy  | yes   | Rte-75  |
| yes   | sunny   | no    | Rte-113 |
| no    | raining | no    | Rte-113 |
| no    | raining | yes   | Rte-75  |
| yes   | cloudy  | yes   | Rte-75  |
| no    | sunny   | no    | Rte-113 |
| yes   | sunny   | no    | Rte-75  |
| no    | cloudy  | no    | Rte-113 |



# Decision Trees Split the Training Set

If  $x_0 = \text{no}$  is chosen, we recurse on both sides.

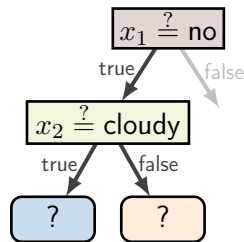
| $x_1$ | $x_2$   | $x_3$ | $y$     |
|-------|---------|-------|---------|
| no    | sunny   | no    | Rte-113 |
| no    | sunny   | yes   | Rte-113 |
| no    | cloudy  | yes   | Rte-75  |
| yes   | sunny   | no    | Rte-113 |
| no    | raining | no    | Rte-113 |
| no    | raining | yes   | Rte-75  |
| yes   | cloudy  | yes   | Rte-75  |
| no    | sunny   | no    | Rte-113 |
| yes   | sunny   | no    | Rte-75  |
| no    | cloudy  | no    | Rte-113 |



# Decision Trees Split the Training Set

If  $x_0 = \text{no}$  is chosen, we recurse on both sides.

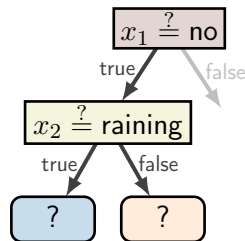
| $x_1$ | $x_2$   | $x_3$ | $y$     |
|-------|---------|-------|---------|
| no    | sunny   | no    | Rte-113 |
| no    | sunny   | yes   | Rte-113 |
| no    | cloudy  | yes   | Rte-75  |
| yes   | sunny   | no    | Rte-113 |
| no    | raining | no    | Rte-113 |
| no    | raining | yes   | Rte-75  |
| yes   | cloudy  | yes   | Rte-75  |
| no    | sunny   | no    | Rte-113 |
| yes   | sunny   | no    | Rte-75  |
| no    | cloudy  | no    | Rte-113 |



# Decision Trees Split the Training Set

If  $x_0 = \text{no}$  is chosen, we recurse on both sides.

| $x_1$ | $x_2$   | $x_3$ | $y$     |
|-------|---------|-------|---------|
| no    | sunny   | no    | Rte-113 |
| no    | sunny   | yes   | Rte-113 |
| no    | cloudy  | yes   | Rte-75  |
| yes   | sunny   | no    | Rte-113 |
| no    | raining | no    | Rte-113 |
| no    | raining | yes   | Rte-75  |
| yes   | cloudy  | yes   | Rte-75  |
| no    | sunny   | no    | Rte-113 |
| yes   | sunny   | no    | Rte-75  |
| no    | cloudy  | no    | Rte-113 |

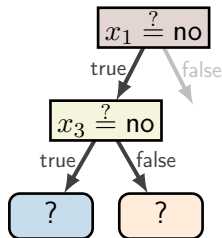




# Decision Trees Split the Training Set

If  $x_0 = \text{no}$  is chosen, we recurse on both sides.

| $x_1$ | $x_2$   | $x_3$ | $y$     |
|-------|---------|-------|---------|
| no    | sunny   | no    | Rte-113 |
| no    | sunny   | yes   | Rte-113 |
| no    | cloudy  | yes   | Rte-75  |
| yes   | sunny   | no    | Rte-113 |
| no    | raining | no    | Rte-113 |
| no    | raining | yes   | Rte-75  |
| yes   | cloudy  | yes   | Rte-75  |
| no    | sunny   | no    | Rte-113 |
| yes   | sunny   | no    | Rte-75  |
| no    | cloudy  | no    | Rte-113 |



# Decision Tree Learning Overview

Given a data set  $X$ , and  $Y$

- If all  $X$  rows are the same, or all  $Y$ s are the same,  
Return tree of a single leaf (the majority class in  $Y$ ).
- Otherwise,
  - ▶ Select the binary test (of  $\vec{x}$ ) that best separates the  $y$ s
  - ▶ Let  $X_t$  and  $Y_t$  be the examples for which the test is true.
  - ▶ Let  $X_f$  and  $Y_f$  be the examples for which the test is false.
  - ▶ Recursively call on  $(X_t, Y_t)$ , assigning result to  $T_t$ .
  - ▶ Recursively call on  $(X_f, Y_f)$ , assigning result to  $T_f$ .
  - ▶ Return tree of binary test, with  $T_t$  on true branch and  $T_f$  on false branch.

# Decision Tree Learning Overview

Given a data set  $X$ , and  $Y$

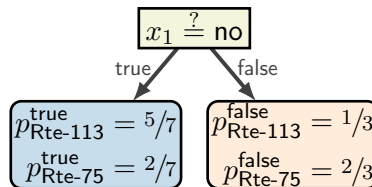
- If all  $X$  rows are the same, or all  $Y$ s are the same,  
Return tree of a single leaf (the majority class in  $Y$ ).
- Otherwise,
  - ▶ Select the binary test (of  $\vec{x}$ ) that **best** separates the  $y$ s
  - ▶ Let  $X_t$  and  $Y_t$  be the examples for which the test is true.
  - ▶ Let  $X_f$  and  $Y_f$  be the examples for which the test is false.
  - ▶ Recursively call on  $(X_t, Y_t)$ , assigning result to  $T_t$ .
  - ▶ Recursively call on  $(X_f, Y_f)$ , assigning result to  $T_f$ .
  - ▶ Return tree of binary test, with  $T_t$  on true branch and  $T_f$  on false branch.

So what is “**best**?”

# Decision Trees Split the Training Set

Picking the root node:

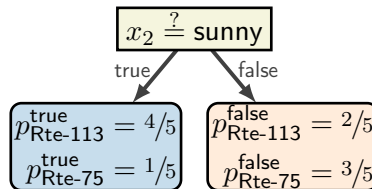
| $x_1$ | $x_2$   | $x_3$ | $y$     |
|-------|---------|-------|---------|
| no    | sunny   | no    | Rte-113 |
| no    | sunny   | yes   | Rte-113 |
| no    | cloudy  | yes   | Rte-75  |
| yes   | sunny   | no    | Rte-113 |
| no    | raining | no    | Rte-113 |
| no    | raining | yes   | Rte-75  |
| yes   | cloudy  | yes   | Rte-75  |
| no    | sunny   | no    | Rte-113 |
| yes   | sunny   | no    | Rte-75  |
| no    | cloudy  | no    | Rte-113 |



# Decision Trees Split the Training Set

Picking the root node:

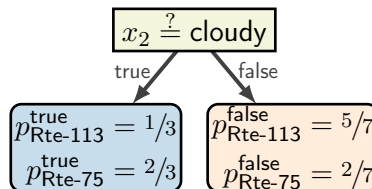
| $x_1$ | $x_2$   | $x_3$ | $y$     |
|-------|---------|-------|---------|
| no    | sunny   | no    | Rte-113 |
| no    | sunny   | yes   | Rte-113 |
| no    | cloudy  | yes   | Rte-75  |
| yes   | sunny   | no    | Rte-113 |
| no    | raining | no    | Rte-113 |
| no    | raining | yes   | Rte-75  |
| yes   | cloudy  | yes   | Rte-75  |
| no    | sunny   | no    | Rte-113 |
| yes   | sunny   | no    | Rte-75  |
| no    | cloudy  | no    | Rte-113 |



# Decision Trees Split the Training Set

Picking the root node:

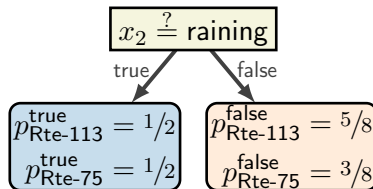
| $x_1$ | $x_2$   | $x_3$ | $y$     |
|-------|---------|-------|---------|
| no    | sunny   | no    | Rte-113 |
| no    | sunny   | yes   | Rte-113 |
| no    | cloudy  | yes   | Rte-75  |
| yes   | sunny   | no    | Rte-113 |
| no    | raining | no    | Rte-113 |
| no    | raining | yes   | Rte-75  |
| yes   | cloudy  | yes   | Rte-75  |
| no    | sunny   | no    | Rte-113 |
| yes   | sunny   | no    | Rte-75  |
| no    | cloudy  | no    | Rte-113 |



# Decision Trees Split the Training Set

Picking the root node:

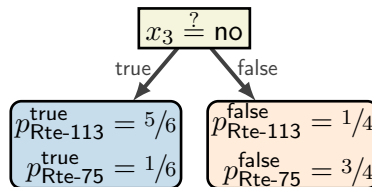
| $x_1$ | $x_2$   | $x_3$ | $y$     |
|-------|---------|-------|---------|
| no    | sunny   | no    | Rte-113 |
| no    | sunny   | yes   | Rte-113 |
| no    | cloudy  | yes   | Rte-75  |
| yes   | sunny   | no    | Rte-113 |
| no    | raining | no    | Rte-113 |
| no    | raining | yes   | Rte-75  |
| yes   | cloudy  | yes   | Rte-75  |
| no    | sunny   | no    | Rte-113 |
| yes   | sunny   | no    | Rte-75  |
| no    | cloudy  | no    | Rte-113 |



# Decision Trees Split the Training Set

Picking the root node:

| $x_1$ | $x_2$   | $x_3$ | $y$     |
|-------|---------|-------|---------|
| no    | sunny   | no    | Rte-113 |
| no    | sunny   | yes   | Rte-113 |
| no    | cloudy  | yes   | Rte-75  |
| yes   | sunny   | no    | Rte-113 |
| no    | raining | no    | Rte-113 |
| no    | raining | yes   | Rte-75  |
| yes   | cloudy  | yes   | Rte-75  |
| no    | sunny   | no    | Rte-113 |
| yes   | sunny   | no    | Rte-75  |
| no    | cloudy  | no    | Rte-113 |

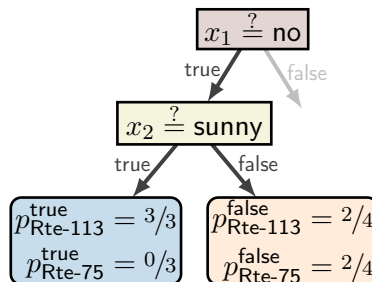




# Decision Trees Split the Training Set

If  $x_0 = \text{no}$  is chosen, we recurse on both sides.

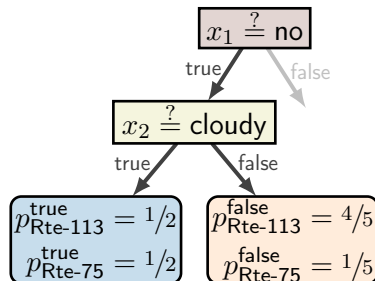
| $x_1$ | $x_2$   | $x_3$ | $y$     |
|-------|---------|-------|---------|
| no    | sunny   | no    | Rte-113 |
| no    | sunny   | yes   | Rte-113 |
| no    | cloudy  | yes   | Rte-75  |
| yes   | sunny   | no    | Rte-113 |
| no    | raining | no    | Rte-113 |
| no    | raining | yes   | Rte-75  |
| yes   | cloudy  | yes   | Rte-75  |
| no    | sunny   | no    | Rte-113 |
| yes   | sunny   | no    | Rte-75  |
| no    | cloudy  | no    | Rte-113 |



# Decision Trees Split the Training Set

If  $x_0 = \text{no}$  is chosen, we recurse on both sides.

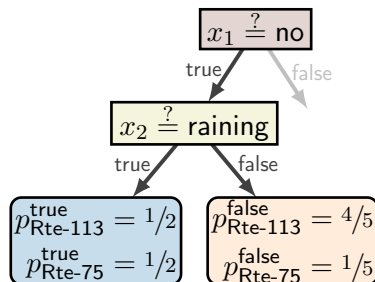
| $x_1$ | $x_2$   | $x_3$ | $y$     |
|-------|---------|-------|---------|
| no    | sunny   | no    | Rte-113 |
| no    | sunny   | yes   | Rte-113 |
| no    | cloudy  | yes   | Rte-75  |
| yes   | sunny   | no    | Rte-113 |
| no    | raining | no    | Rte-113 |
| no    | raining | yes   | Rte-75  |
| yes   | cloudy  | yes   | Rte-75  |
| no    | sunny   | no    | Rte-113 |
| yes   | sunny   | no    | Rte-75  |
| no    | cloudy  | no    | Rte-113 |



# Decision Trees Split the Training Set

If  $x_0 = \text{no}$  is chosen, we recurse on both sides.

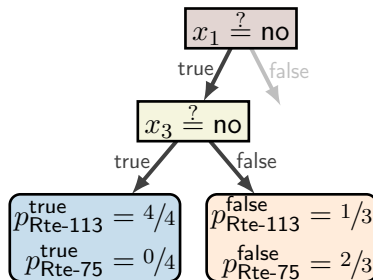
| $x_1$ | $x_2$   | $x_3$ | $y$     |
|-------|---------|-------|---------|
| no    | sunny   | no    | Rte-113 |
| no    | sunny   | yes   | Rte-113 |
| no    | cloudy  | yes   | Rte-75  |
| yes   | sunny   | no    | Rte-113 |
| no    | raining | no    | Rte-113 |
| no    | raining | yes   | Rte-75  |
| yes   | cloudy  | yes   | Rte-75  |
| no    | sunny   | no    | Rte-113 |
| yes   | sunny   | no    | Rte-75  |
| no    | cloudy  | no    | Rte-113 |



# Decision Trees Split the Training Set

If  $x_0 = \text{no}$  is chosen, we recurse on both sides.

| $x_1$ | $x_2$   | $x_3$ | $y$     |
|-------|---------|-------|---------|
| no    | sunny   | no    | Rte-113 |
| no    | sunny   | yes   | Rte-113 |
| no    | cloudy  | yes   | Rte-75  |
| yes   | sunny   | no    | Rte-113 |
| no    | raining | no    | Rte-113 |
| no    | raining | yes   | Rte-75  |
| yes   | cloudy  | yes   | Rte-75  |
| no    | sunny   | no    | Rte-113 |
| yes   | sunny   | no    | Rte-75  |
| no    | cloudy  | no    | Rte-113 |



# Scoring function

For any potential split, we calculate

$$\begin{bmatrix} p_1^{\text{true}} \\ p_2^{\text{true}} \\ \vdots \\ p_c^{\text{true}} \end{bmatrix}$$

$$\begin{bmatrix} p_1^{\text{false}} \\ p_2^{\text{false}} \\ \vdots \\ p_c^{\text{false}} \end{bmatrix}$$

# Scoring function

For any potential split, we calculate

$$\vec{p}^{\text{true}} = \begin{bmatrix} p_1^{\text{true}} \\ p_2^{\text{true}} \\ \vdots \\ p_c^{\text{true}} \end{bmatrix}$$

$$\vec{p}^{\text{false}} = \begin{bmatrix} p_1^{\text{false}} \\ p_2^{\text{false}} \\ \vdots \\ p_c^{\text{false}} \end{bmatrix}$$

# Scoring function

For any potential split, we calculate

$$\vec{p}^{\text{true}} = \begin{bmatrix} p_1^{\text{true}} \\ p_2^{\text{true}} \\ \vdots \\ p_c^{\text{true}} \end{bmatrix} \qquad \vec{p}^{\text{false}} = \begin{bmatrix} p_1^{\text{false}} \\ p_2^{\text{false}} \\ \vdots \\ p_c^{\text{false}} \end{bmatrix}$$

$$\text{score} = \underbrace{m^{\text{true}}}_{\# \text{ true ex}} G(\vec{p}^{\text{true}}) + \underbrace{m^{\text{false}}}_{\# \text{ false ex}} G(\vec{p}^{\text{false}})$$

$G$  is an impurity measure.

# Scoring function

For any potential split, we calculate

$$\vec{p}^{\text{true}} = \begin{bmatrix} p_1^{\text{true}} \\ p_2^{\text{true}} \\ \vdots \\ p_c^{\text{true}} \end{bmatrix} \qquad \vec{p}^{\text{false}} = \begin{bmatrix} p_1^{\text{false}} \\ p_2^{\text{false}} \\ \vdots \\ p_c^{\text{false}} \end{bmatrix}$$

$$\text{score} = \underbrace{m^{\text{true}}}_{\# \text{ true ex}} G(\vec{p}^{\text{true}}) + \underbrace{m^{\text{false}}}_{\# \text{ false ex}} G(\vec{p}^{\text{false}})$$

$G$  is an impurity measure.

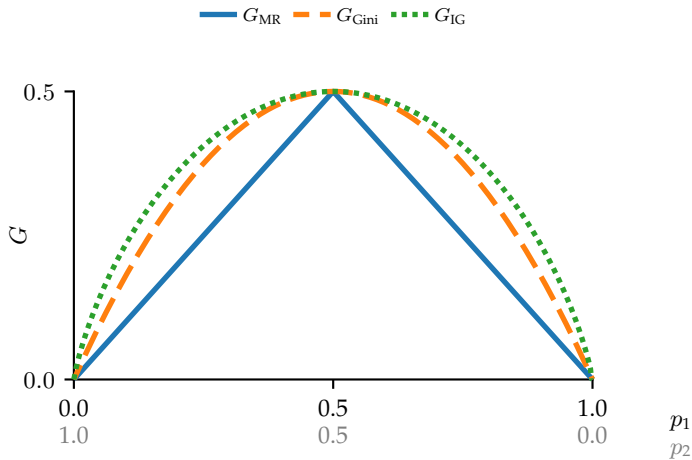
(we try this for all possible splits and pick the lowest value)



# Impurity Measures

| Name                   |                              | general                                   | binary classification          |
|------------------------|------------------------------|-------------------------------------------|--------------------------------|
| misclassification rate | $G_{\text{MR}}(\vec{p}) =$   | $1 - \max_{c \in \{1, 2, \dots, C\}} p_c$ | $= \min(p_1, p_2)$             |
| Gini index             | $G_{\text{Gini}}(\vec{p}) =$ | $1 - \sum_{c=1}^C p_c^2$                  | $= 2p_1p_2$                    |
| information gain       | $G_{\text{IG}}(\vec{p}) =$   | $-\sum_{c=1}^C p_c \lg p_c$               | $= -p_1 \lg p_1 - p_2 \lg p_2$ |

# Impurity Measures



# Continuous Features

What about features that are not categorical?

# Continuous Features

What about features that are not categorical?

If  $x_1$  is a real-valued feature, we consider tests of the form

$$x_1 < t$$

for different values of  $t$ .

# Continuous Features

What about features that are not categorical?

If  $x_1$  is a real-valued feature, we consider tests of the form

$$x_1 < t$$

for different values of  $t$ .

| feature type | number values | number tests |
|--------------|---------------|--------------|
| categorical  | $k$           | $k$          |
| continuous   | $\infty$      | ???          |

# Continuous Features

What about features that are not categorical?

If  $x_1$  is a real-valued feature, we consider tests of the form

$$x_1 < t$$

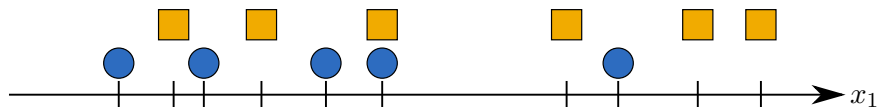
for different values of  $t$ .

| feature type | number values | number tests |
|--------------|---------------|--------------|
| categorical  | $k$           | $k$          |
| continuous   | $\infty$      | $\leq m$     |

# Continuous Features

Consider splitting the dataset on feature  $x_1$  (for instance).

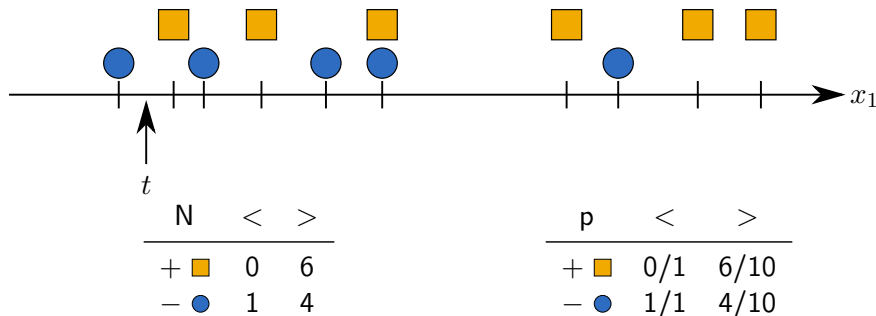
Plotting only  $x_1$  versus  $y$ :



# Continuous Features

Consider splitting the dataset on feature  $x_1$  (for instance).

Plotting only  $x_1$  versus  $y$ :

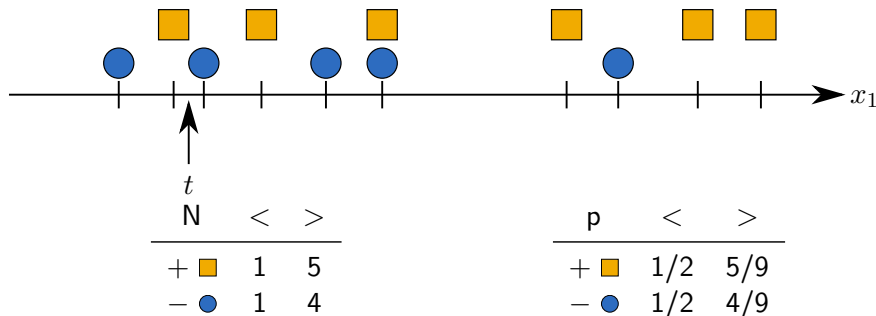




# Continuous Features

Consider splitting the dataset on feature  $x_1$  (for instance).

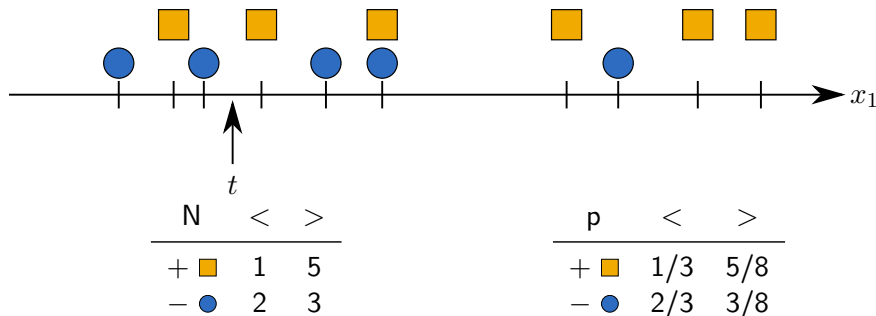
Plotting only  $x_1$  versus  $y$ :



# Continuous Features

Consider splitting the dataset on feature  $x_1$  (for instance).

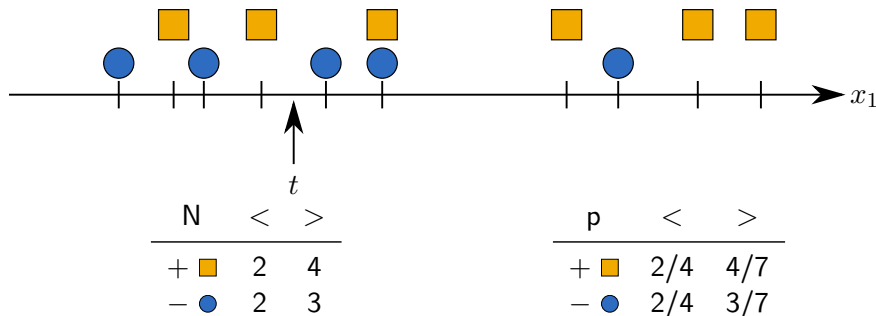
Plotting only  $x_1$  versus  $y$ :



# Continuous Features

Consider splitting the dataset on feature  $x_1$  (for instance).

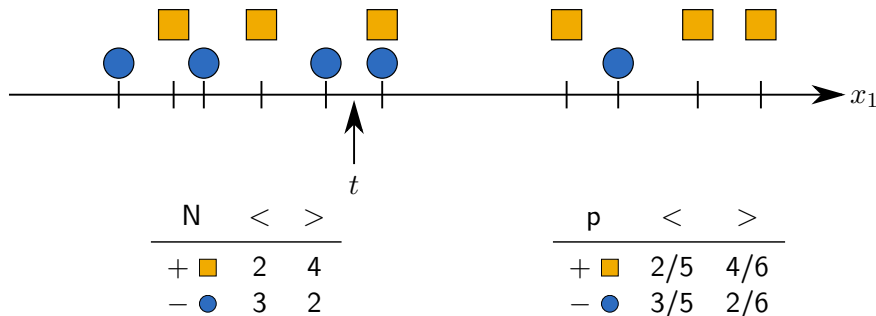
Plotting only  $x_1$  versus  $y$ :



# Continuous Features

Consider splitting the dataset on feature  $x_1$  (for instance).

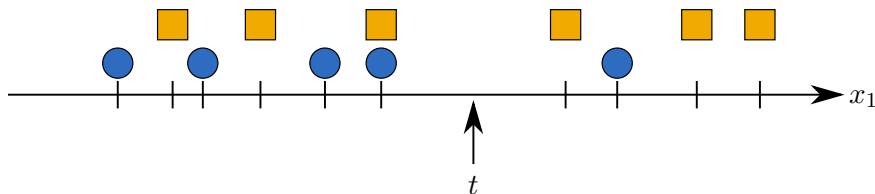
Plotting only  $x_1$  versus  $y$ :



# Continuous Features

Consider splitting the dataset on feature  $x_1$  (for instance).

Plotting only  $x_1$  versus  $y$ :



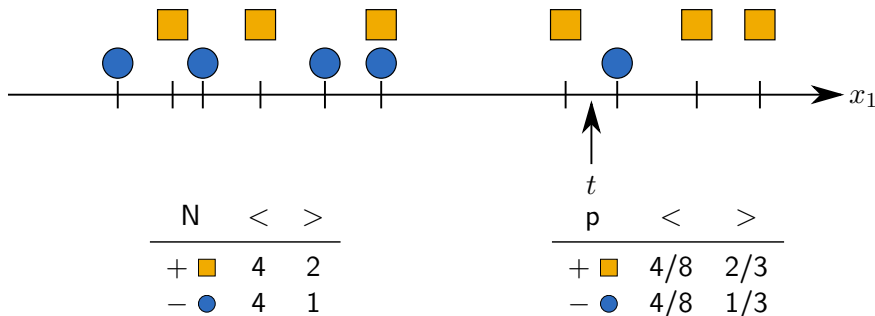
| N |   | < | > |
|---|---|---|---|
| + | ■ | 3 | 3 |
| - | ● | 4 | 1 |

| p |   | <   | >   |
|---|---|-----|-----|
| + | ■ | 3/7 | 3/4 |
| - | ● | 4/7 | 1/4 |

# Continuous Features

Consider splitting the dataset on feature  $x_1$  (for instance).

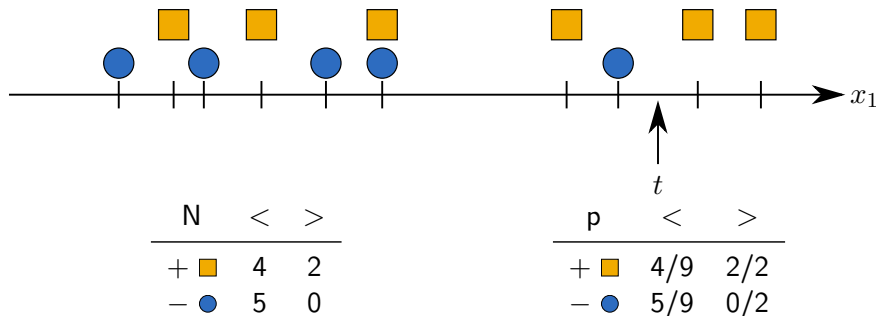
Plotting only  $x_1$  versus  $y$ :



# Continuous Features

Consider splitting the dataset on feature  $x_1$  (for instance).

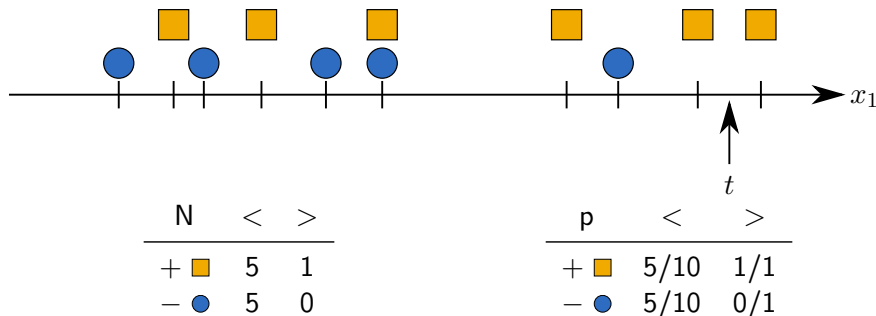
Plotting only  $x_1$  versus  $y$ :



# Continuous Features

Consider splitting the dataset on feature  $x_1$  (for instance).

Plotting only  $x_1$  versus  $y$ :

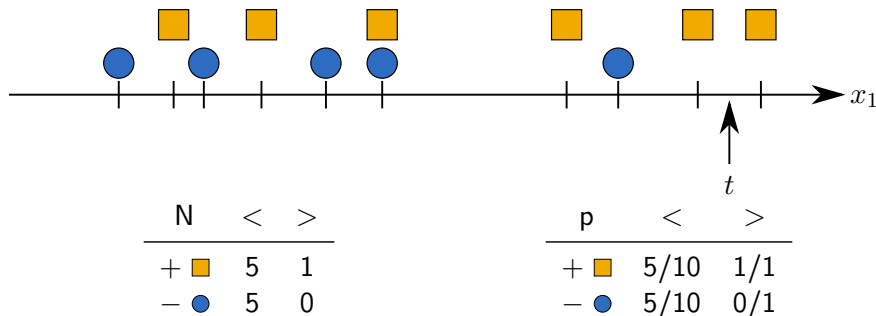




# Continuous Features

Consider splitting the dataset on feature  $x_1$  (for instance).

Plotting only  $x_1$  versus  $y$ :

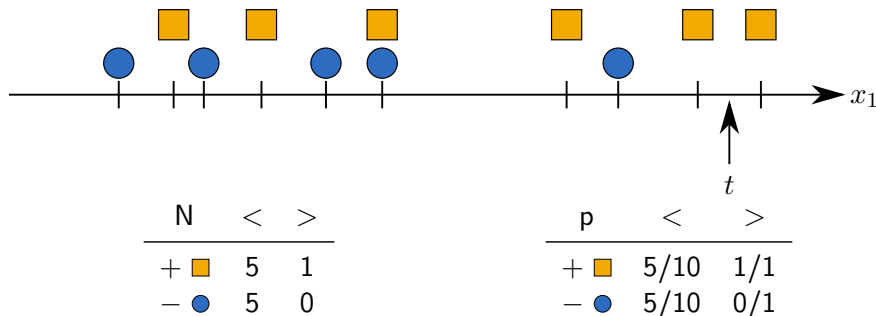


So, with  $n_c$  categorical features, each with  $k$  values and  $n_r$  real-valued features, how many tests must the algorithm check for each split?

# Continuous Features

Consider splitting the dataset on feature  $x_1$  (for instance).

Plotting only  $x_1$  versus  $y$ :

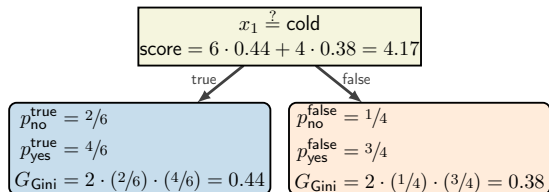


So, with  $n_c$  categorical features, each with  $k$  values and  $n_r$  real-valued features, how many tests must the algorithm check for each split?

Answer:  $n_c \times k + n_r \times m$  for a total running time of  $O(n_c(k + m) + n_r m \ln m)$

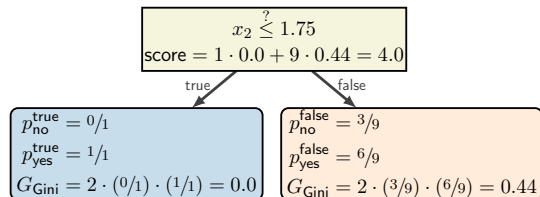
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



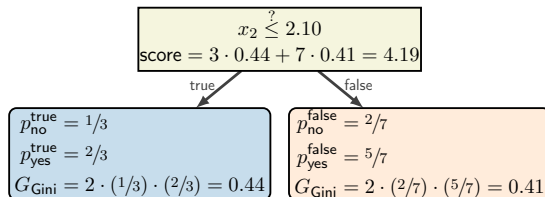
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



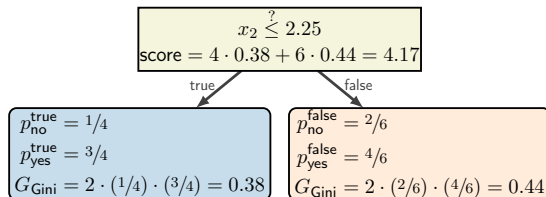
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



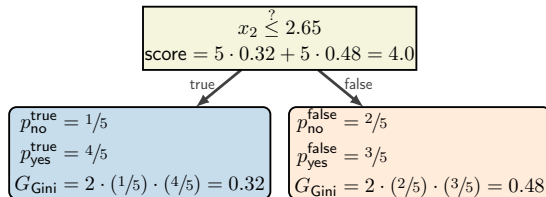
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



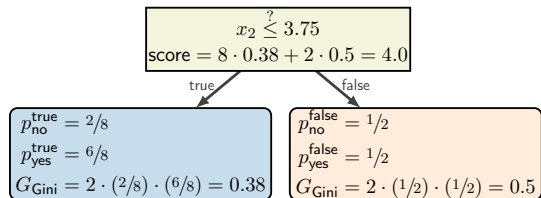
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



# Growing a toy decision tree

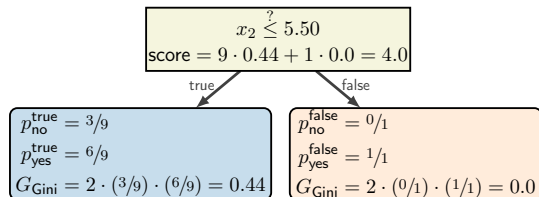
| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |





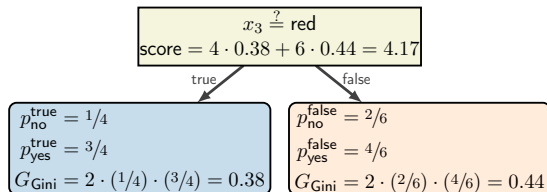
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



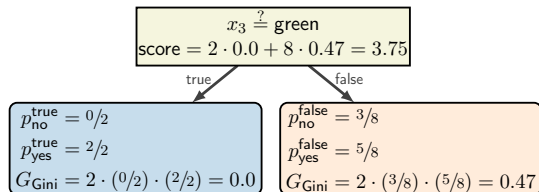
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



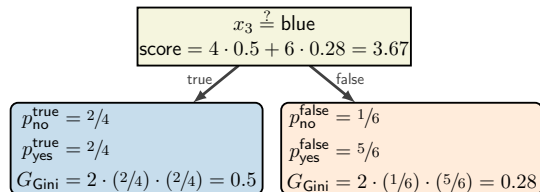
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



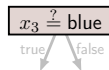
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



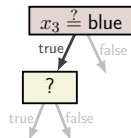
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



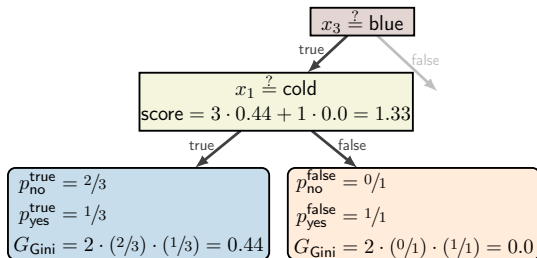
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



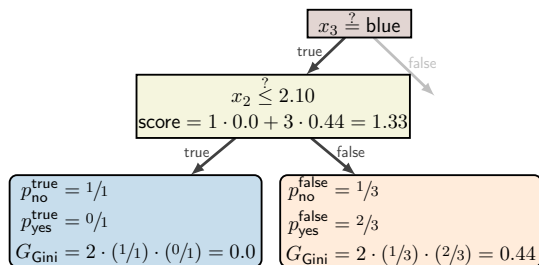
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



# Growing a toy decision tree

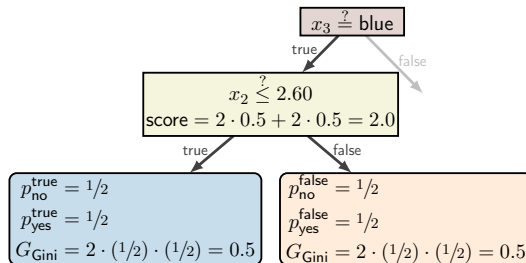
| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |





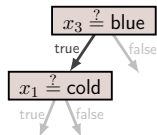
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



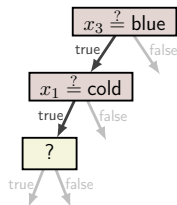
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



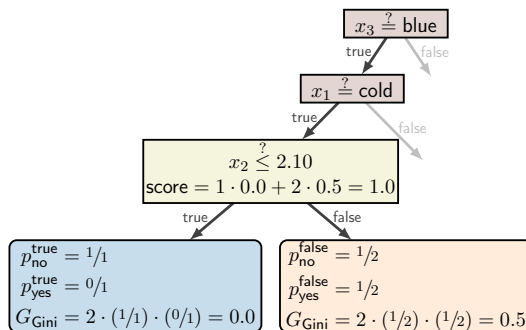
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



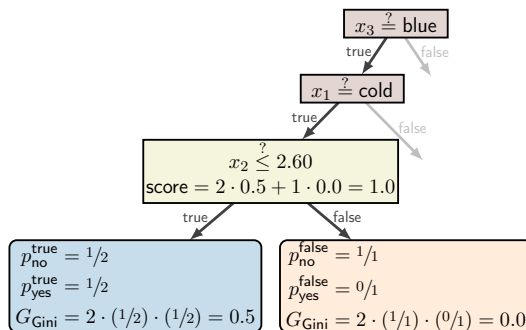
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



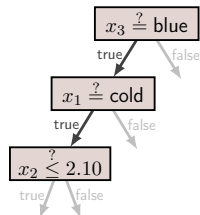
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



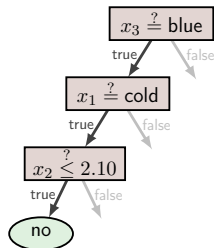
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



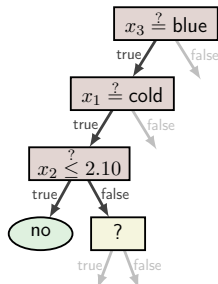
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



# Growing a toy decision tree

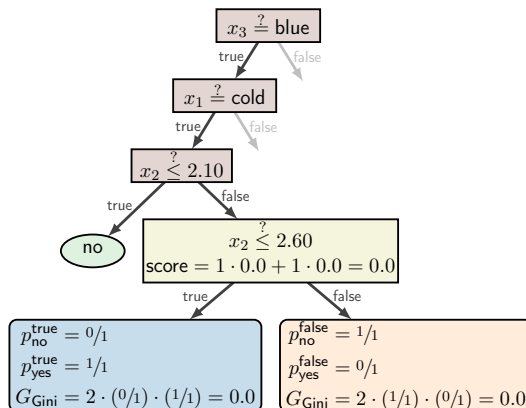
| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |





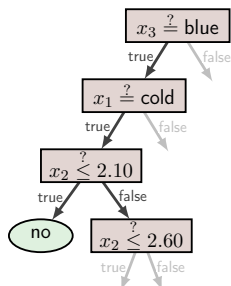
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



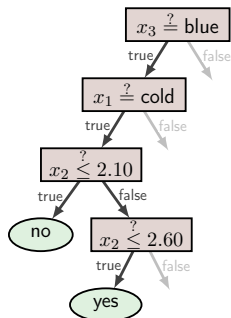
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



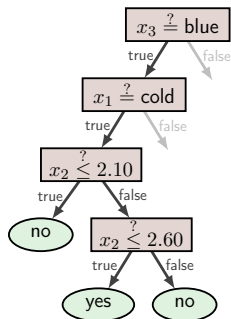
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



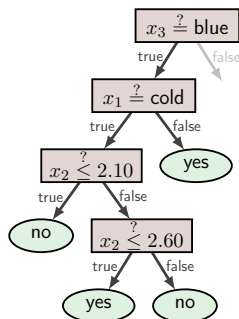
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



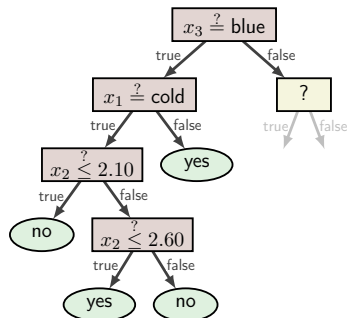
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



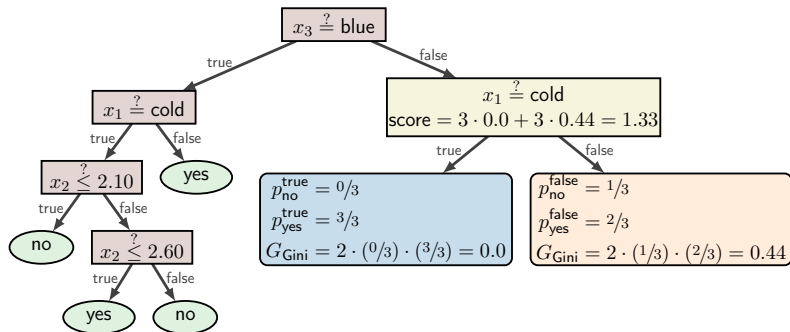
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



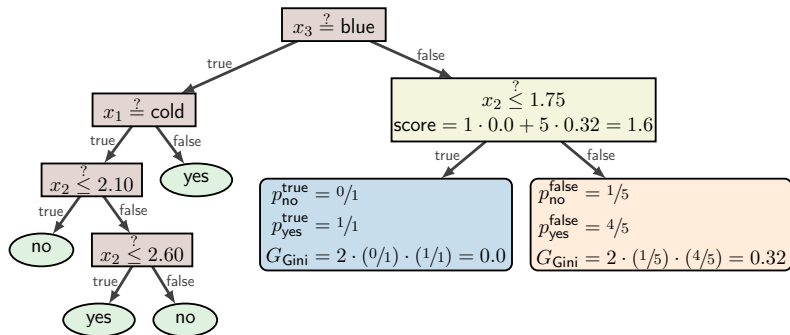
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



# Growing a toy decision tree

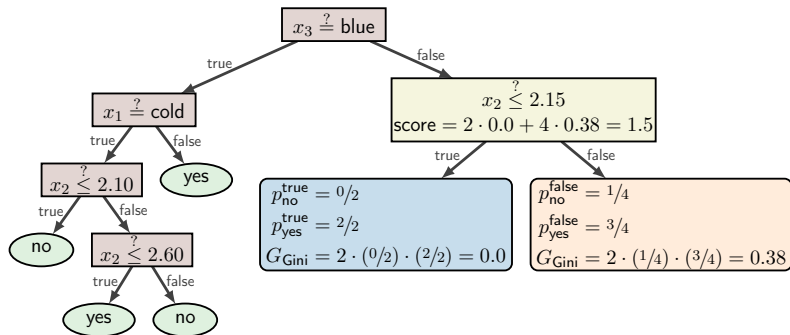
| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |





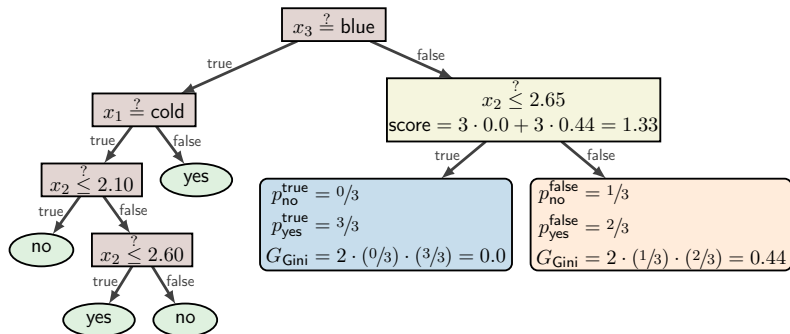
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



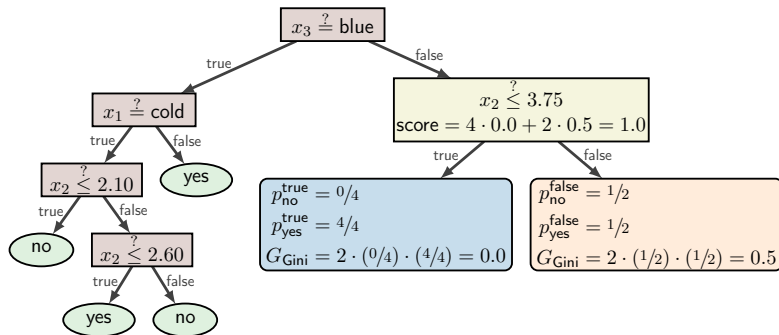
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



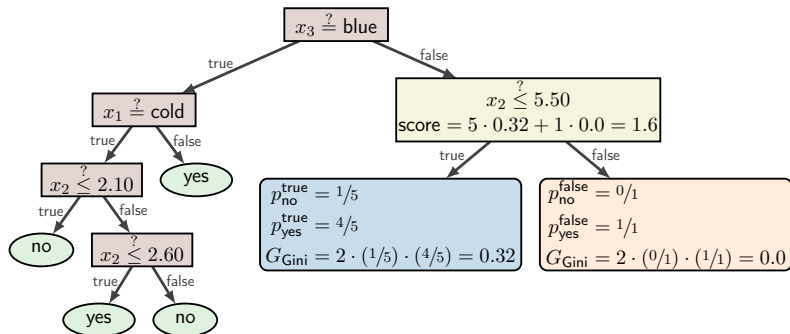
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



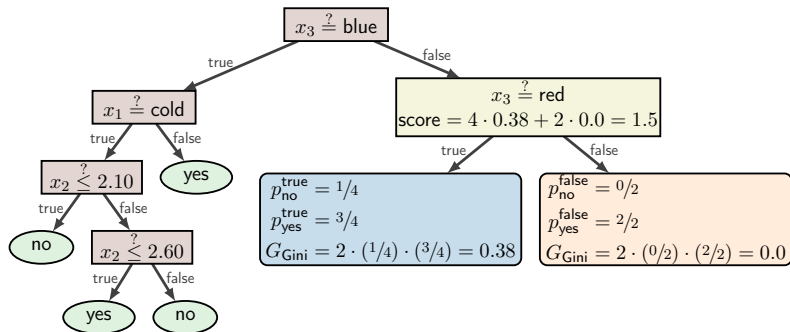
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



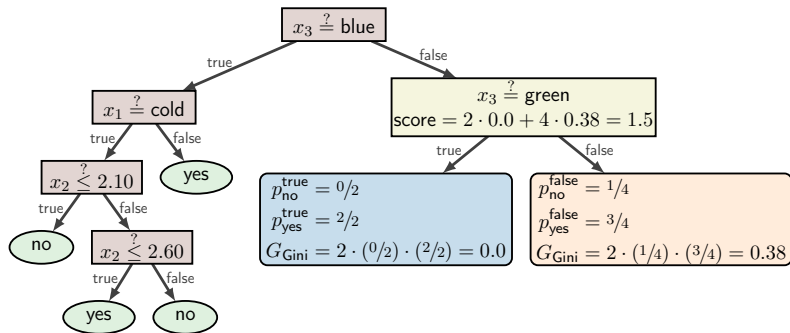
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



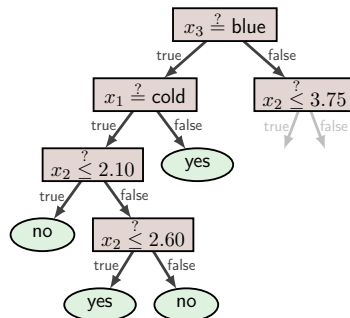
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



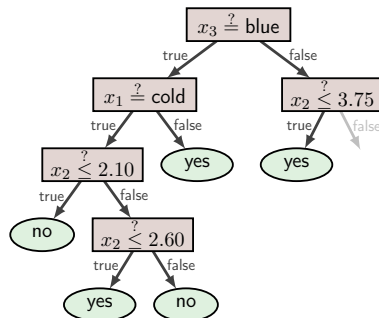
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



# Growing a toy decision tree

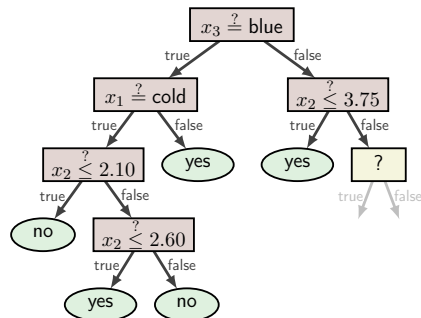
| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |





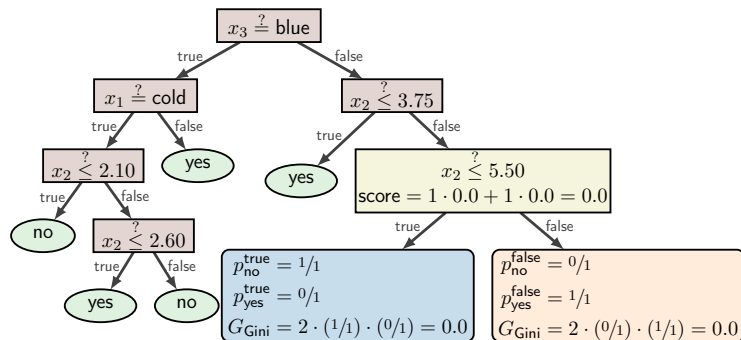
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



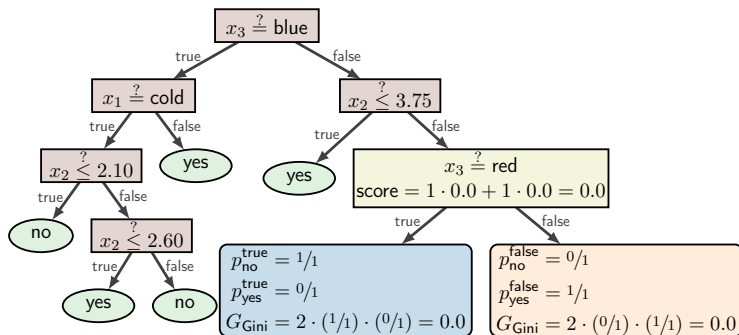
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



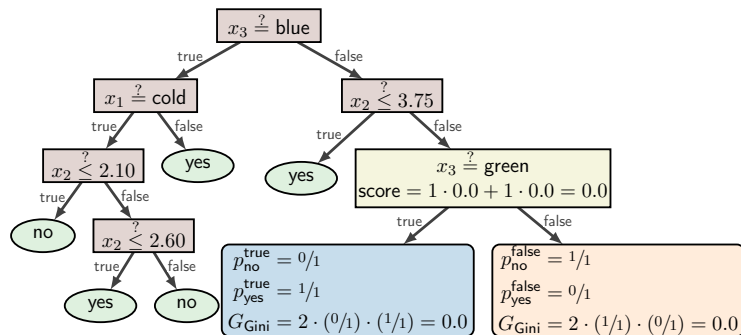
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



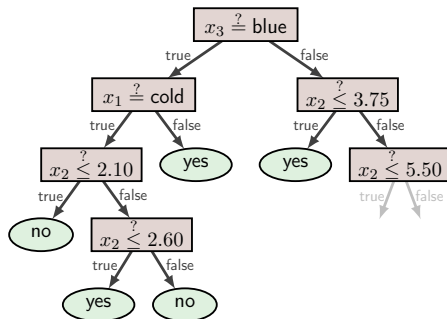
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



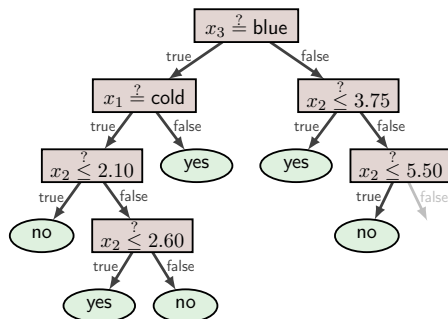
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



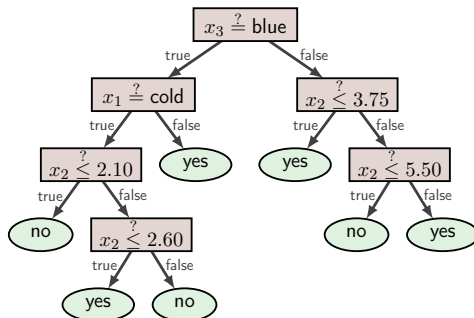
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



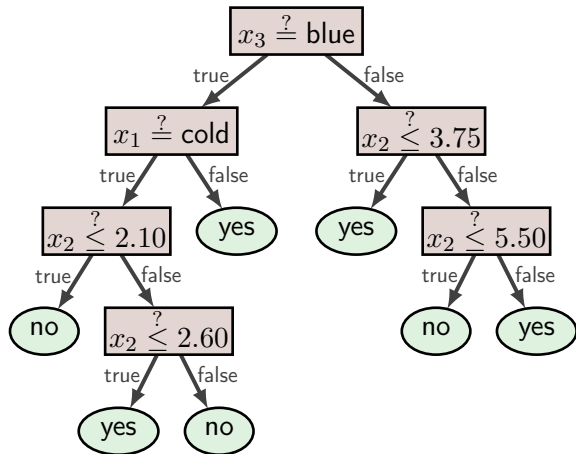
# Growing a toy decision tree

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |



# Decision Tree Pruning

Building a tree like this almost certainly overfits!  
We need to figure out where to stop.

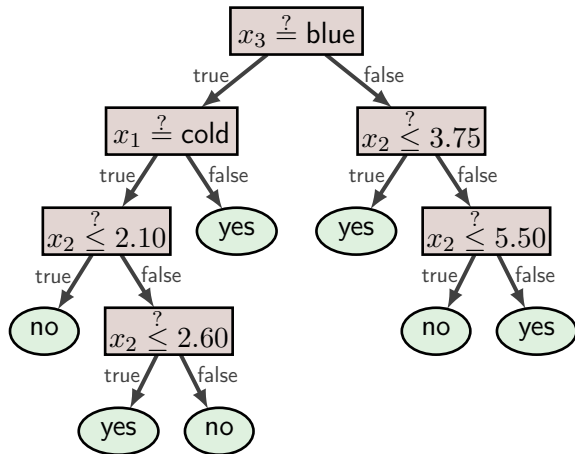




# Decision Tree Pruning

Building a tree like this almost certainly overfits!  
We need to figure out where to stop.

Any subtree (with the same root) could have been the learning result if we chose to stop at a different point.



# Decision Tree Pruning

Consider (recursively, bottom-up) replacing each subtree with a leaf

- Cost-complexity pruning: Prune if it improves  $L + \lambda \|\mathcal{T}\|$  ( $\lambda$  chosen by cross-validation)
- Chi-squared pruning: Use statistical test to check if test is correlated with label
- Reduced error pruning: Prune if doing so does not change or improves pruning set (like a validation set) error

# Decision Tree Pruning

Consider (recursively, bottom-up) replacing each subtree with a leaf

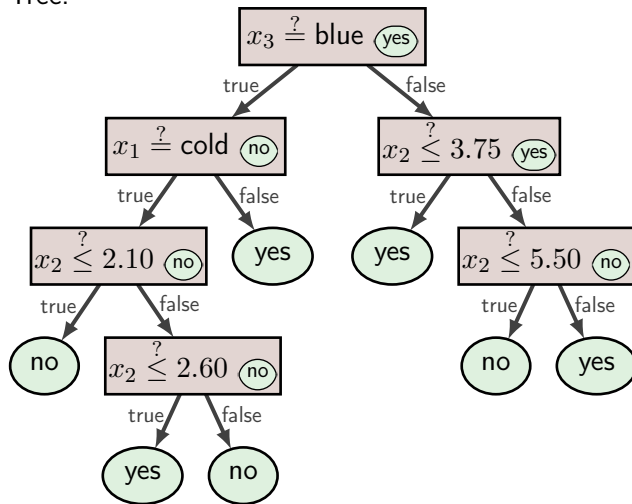
- Cost-complexity pruning: Prune if it improves  $L + \lambda \|\mathcal{T}\|$  ( $\lambda$  chosen by cross-validation)
- Chi-squared pruning: Use statistical test to check if test is correlated with label
- **Reduced error pruning**: Prune if doing so does not change or improves pruning set (like a validation set) error
  - ▶ Pruned leaf's  $y$  values chosen from training data
  - ▶ Pruned bottom-up

# Pruning a toy decision tree

Training Data:

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 2.0   | red   | yes |
| cold  | 2.3   | green | yes |
| hot   | 4.5   | red   | no  |
| hot   | 6.5   | green | yes |
| cold  | 3.0   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 2.0   | blue  | no  |
| cold  | 2.2   | blue  | yes |
| hot   | 3.0   | red   | yes |
| cold  | 1.5   | red   | yes |

Tree:

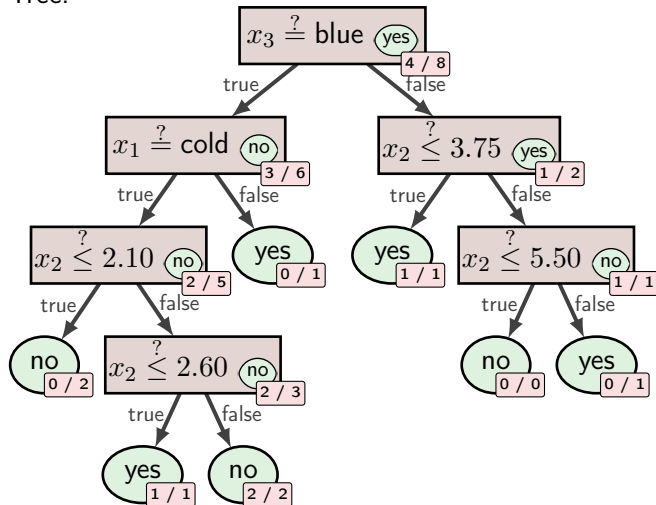


# Pruning a toy decision tree

Pruning (validation) Data:

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 3.5   | blue  | yes |
| cold  | 1.5   | blue  | no  |
| cold  | 2.0   | blue  | no  |
| hot   | 3.5   | red   | no  |
| cold  | 5.9   | green | yes |
| cold  | 2.2   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 3.0   | blue  | yes |

Tree:

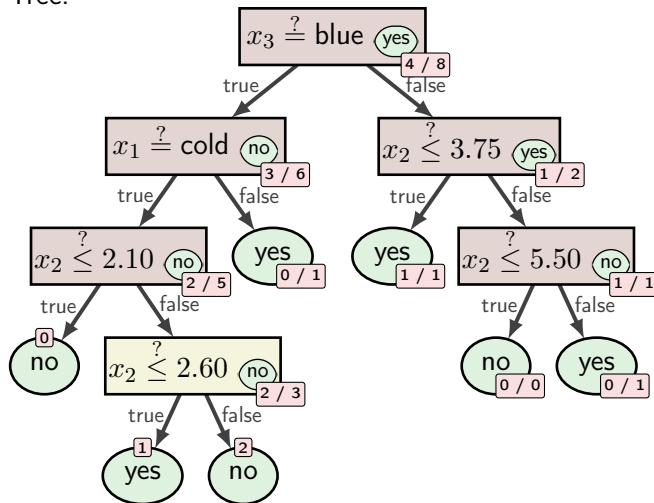


# Pruning a toy decision tree

Pruning (validation) Data:

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 3.5   | blue  | yes |
| cold  | 1.5   | blue  | no  |
| cold  | 2.0   | blue  | no  |
| hot   | 3.5   | red   | no  |
| cold  | 5.9   | green | yes |
| cold  | 2.2   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 3.0   | blue  | yes |

Tree:

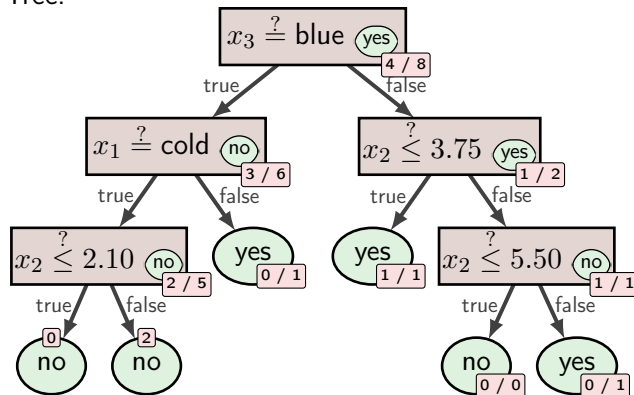


# Pruning a toy decision tree

Pruning (validation) Data:

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 3.5   | blue  | yes |
| cold  | 1.5   | blue  | no  |
| cold  | 2.0   | blue  | no  |
| hot   | 3.5   | red   | no  |
| cold  | 5.9   | green | yes |
| cold  | 2.2   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 3.0   | blue  | yes |

Tree:

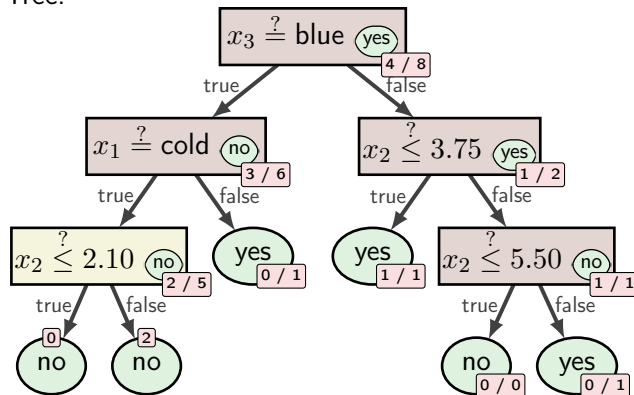


# Pruning a toy decision tree

Pruning (validation) Data:

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 3.5   | blue  | yes |
| cold  | 1.5   | blue  | no  |
| cold  | 2.0   | blue  | no  |
| hot   | 3.5   | red   | no  |
| cold  | 5.9   | green | yes |
| cold  | 2.2   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 3.0   | blue  | yes |

Tree:



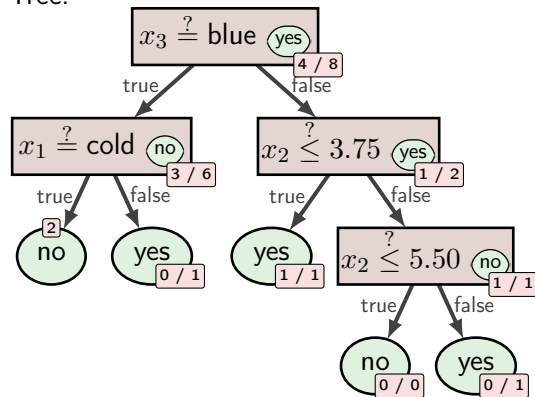


# Pruning a toy decision tree

Pruning (validation) Data:

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 3.5   | blue  | yes |
| cold  | 1.5   | blue  | no  |
| cold  | 2.0   | blue  | no  |
| hot   | 3.5   | red   | no  |
| cold  | 5.9   | green | yes |
| cold  | 2.2   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 3.0   | blue  | yes |

Tree:

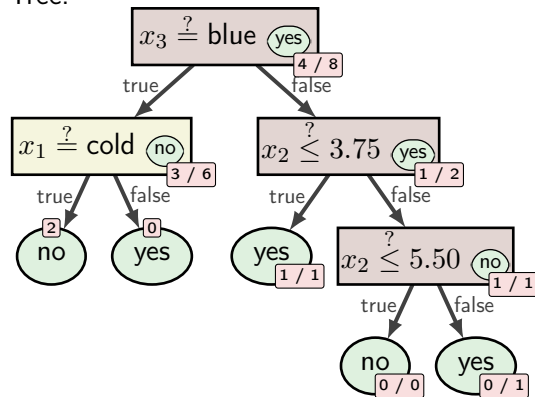


# Pruning a toy decision tree

Pruning (validation) Data:

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 3.5   | blue  | yes |
| cold  | 1.5   | blue  | no  |
| cold  | 2.0   | blue  | no  |
| hot   | 3.5   | red   | no  |
| cold  | 5.9   | green | yes |
| cold  | 2.2   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 3.0   | blue  | yes |

Tree:

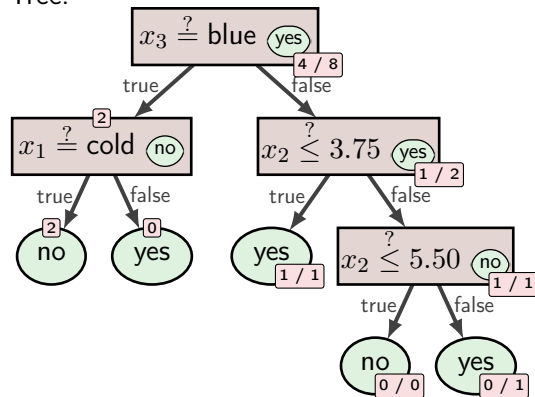


# Pruning a toy decision tree

Pruning (validation) Data:

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 3.5   | blue  | yes |
| cold  | 1.5   | blue  | no  |
| cold  | 2.0   | blue  | no  |
| hot   | 3.5   | red   | no  |
| cold  | 5.9   | green | yes |
| cold  | 2.2   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 3.0   | blue  | yes |

Tree:

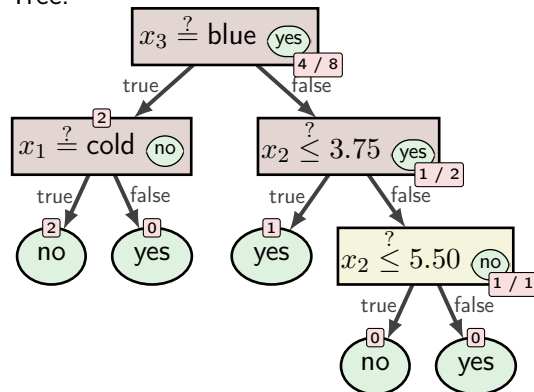


# Pruning a toy decision tree

Pruning (validation) Data:

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 3.5   | blue  | yes |
| cold  | 1.5   | blue  | no  |
| cold  | 2.0   | blue  | no  |
| hot   | 3.5   | red   | no  |
| cold  | 5.9   | green | yes |
| cold  | 2.2   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 3.0   | blue  | yes |

Tree:

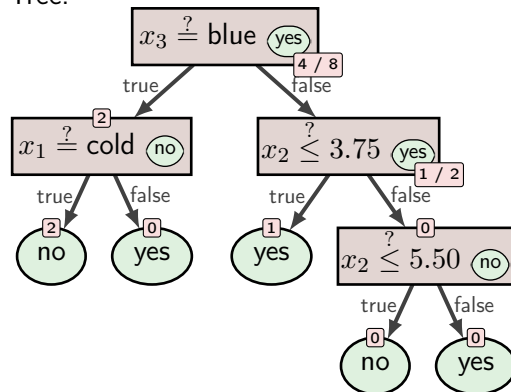


# Pruning a toy decision tree

Pruning (validation) Data:

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 3.5   | blue  | yes |
| cold  | 1.5   | blue  | no  |
| cold  | 2.0   | blue  | no  |
| hot   | 3.5   | red   | no  |
| cold  | 5.9   | green | yes |
| cold  | 2.2   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 3.0   | blue  | yes |

Tree:

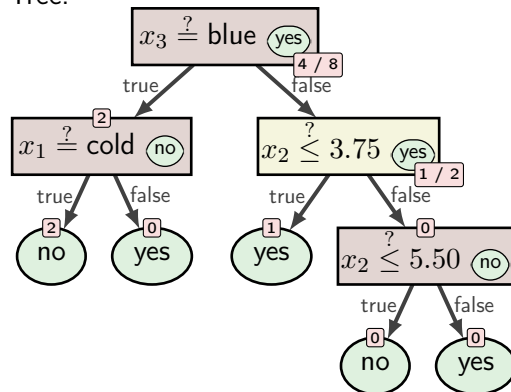


# Pruning a toy decision tree

Pruning (validation) Data:

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 3.5   | blue  | yes |
| cold  | 1.5   | blue  | no  |
| cold  | 2.0   | blue  | no  |
| hot   | 3.5   | red   | no  |
| cold  | 5.9   | green | yes |
| cold  | 2.2   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 3.0   | blue  | yes |

Tree:

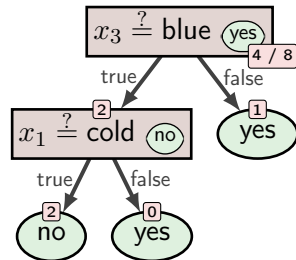


# Pruning a toy decision tree

Pruning (validation) Data:

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 3.5   | blue  | yes |
| cold  | 1.5   | blue  | no  |
| cold  | 2.0   | blue  | no  |
| hot   | 3.5   | red   | no  |
| cold  | 5.9   | green | yes |
| cold  | 2.2   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 3.0   | blue  | yes |

Tree:

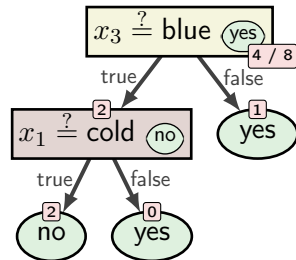


# Pruning a toy decision tree

Pruning (validation) Data:

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 3.5   | blue  | yes |
| cold  | 1.5   | blue  | no  |
| cold  | 2.0   | blue  | no  |
| hot   | 3.5   | red   | no  |
| cold  | 5.9   | green | yes |
| cold  | 2.2   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 3.0   | blue  | yes |

Tree:



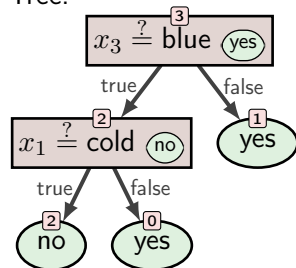


# Pruning a toy decision tree

Pruning (validation) Data:

| $x_1$ | $x_2$ | $x_3$ | $y$ |
|-------|-------|-------|-----|
| cold  | 3.5   | blue  | yes |
| cold  | 1.5   | blue  | no  |
| cold  | 2.0   | blue  | no  |
| hot   | 3.5   | red   | no  |
| cold  | 5.9   | green | yes |
| cold  | 2.2   | blue  | no  |
| hot   | 3.0   | blue  | yes |
| cold  | 3.0   | blue  | yes |

Tree:



# Decision Tree Variants

- Non-binary splits: not usually done (insufficient data)

# Decision Tree Variants

- Non-binary splits: not usually done (insufficient data)
- Categorical features test of the form  $x_i \in \{v_1, v_2, \dots\}$ :  
Instead of asking  $x_i = v$ , ask if one of a set of values.  
Many different splits, but can build up set incrementally.

# Decision Tree Variants

- Non-binary splits: not usually done (insufficient data)
- Categorical features test of the form  $x_i \in \{v_1, v_2, \dots\}$ :  
Instead of asking  $x_i = v$ , ask if one of a set of values.  
Many different splits, but can build up set incrementally.
- Missing values:
  - ▶ Use ternary splits (extra “missing” branch)
  - ▶ Use surrogate splits (a second split that best mimics the primary split)

# Decision Trees

## Benefits:

- Interpretable (?)
- Missing values
- Categorical features

## Problems:

- High variance
- Not smooth
- Cannot produce linear model

# Spam Example

From: cheapsales@buystufffromme.com  
To: ang@cs.stanford.edu  
Subject: Buy now!

Deal of the week! Buy now!  
Rolex w4tchs - \$100  
Medicine (any kind) - \$50  
Also low cost M0rgages  
available.

Spam

From: Alfred Ng  
To: ang@cs.stanford.edu  
Subject: Christmas dates?

Hey Andrew,  
Was talking to Mom about plans  
for Xmas. When do you get off  
work. Meet Dec 22?  
Alf

Non-spam

---

Top image courtesy of Andrew Ng  
Bottom example from George Forman of HP Labs

# Spam Example

From: cheapsales@buystufffromme.com  
To: ang@cs.stanford.edu  
Subject: Buy now!

Deal of the week! Buy now!  
Rolex watches - \$100  
Medicine (any kind) - \$50  
Also low cost M0rgages  
available.

Spam

From: Alfred Ng  
To: ang@cs.stanford.edu  
Subject: Christmas dates?

Hey Andrew,  
Was talking to Mom about plans  
for Xmas. When do you get off  
work. Meet Dec 22?  
Alf

Non-spam

Convert e-mail into features:

- (1) CAPMAX: length of longest uninterrupted sequence of capitals
- (1) CAPAVE: average length of capitals sequences
- (1) CAPTOT: total number of capitals
- (48) % of words that match a particular word, ex: business, address, internet, free, george
- (6) % of characters that match ; ( [ ! \$ #

---

Top image courtesy of Andrew Ng

Bottom example from George Forman of HP Labs

# Spam Example

